Yanshuai Qin

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Profile

PhD in Mathematics from UC Berkeley with postdoctoral research experience in Germany, specializing in arithmetic geometry. Strong background in probability, statistics, and numerical methods, with hands-on experience implementing deep learning models and analyzing complex datasets. Combines rigorous analytical thinking with strong quantitative problem-solving skills.

Technical Skills

Machine Learning	PyTorch, TensorFlow, scikit-learn, LLM pretraining, supervised fine-tuning, RLHF, quantization, evaluation, deployment
Programming	Python, C++, Git/GitHub, Linux
Data & Scientific Computing	NumPy, pandas, SciPy, Jupyter
Mathematics	Probability, statistics, numerical analysis, optimization
Languages	Chinese (native), English (fluent)

Selected Projects

- **Nanochat End-to-End LLM Pipeline** (GitHub): Built and validated an end-to-end LLM pipeline on Kaggle using the `nanochat` codebase, covering tokenizer training, pretraining, supervised fine-tuning, distillation, DPO/PPO/RL post-training, quantization, evaluation, and deployment. Adapted the workflow to run on 2× T4 GPUs by debugging distributed execution, stabilizing data loading in notebook environments, and producing working checkpoints and evaluation outputs at each stage.
- **Cost-Aware Limit Order Book Forecasting** (GitHub): Developed an end-to-end pipeline for short-horizon limit-order-book forecasting and benchmarked XGBoost, CNN-LSTM, and Transformer models using leakage-safe walk-forward validation. Built a transaction-cost-aware backtesting framework incorporating spread, fees, slippage, calibration, and latency.

Experience

Postdoctoral Researcher (Supervisor: Moritz Kerz) *Nov 2023 – Dec 2025*
University of Regensburg, Germany

- Jointly with a mathematician from IMJ-PRG, completely solved a difficult problem raised by Skorobogatov and Zarhin a decade ago; invited to speak at EPFL and IRMA, and selected to present the work at 2025 SRI in Algebraic Geometry, a leading once-a-decade conference.
- Completed, with an IBM cryptography scientist and an expert in p-adic methods, a full generalization of the Artin-Grothendieck theorem (1968) to arbitrary dimension, building on a collaborator's breakthrough on the 30-year-old Berthelot conjecture; the work was presented by the collaborator at an IHES conference.

Visiting Scholar *Oct 2023 – Nov 2023*
Beijing International Center for Mathematical Research, Beijing

Graduate Student Instructor and Researcher *Aug 2018 – Sep 2023*
University of California, Berkeley

- Taught courses in numerical analysis/optimization, probability, statistics, and differential geometry.
- Conducted research in arithmetic geometry, resulting in 5 publications and multiple invited talks at international conferences and institutes, including PRIMA Congress 2022.

Education

PhD, Mathematics, University of California, Berkeley (Advisor: Xinyi Yuan) 2018–2023
MSc, Mathematics, Nanjing University 2016–2018
BSc, Mathematics, Nanjing University 2011–2015

Awards

- Lehmer Fellowship in Number Theory (Spring 2022); UC Berkeley Summer Grants (2020–2023).
- UC Berkeley Conference Travel Grant.